

Phase 4 of 4

LLM Assistant, Security Layers & Notifications

Maternal Health Platform — Development Roadmap · Derived strictly from the Complete Data Flow and Final Maternal LLM Guide documents

What This Phase Covers

This phase builds the patient-only AI lifestyle assistant, the security layers that protect it, and the platform-wide notification system for patient, doctor, and hospital.

NOTE: Single-model architecture for this build: every patient chat message is handled by ONE LLM — openai/gpt 120B, called via the Groq API. There is no second local model and no complexity-based routing agent deciding which model to use. Every message that reaches the LLM step follows the same single pipeline: PII Anonymizer → Prompt Injection Defender → RAG retrieval → openai/gpt 120B via Groq → Response Filter → patient chat window.

1. The Golden Rule

The LLM communicates ONLY with the patient. It has NO direct link to the doctor whatsoever. It does not send notifications, alerts, or messages to the doctor at any point. If something requires a doctor, the LLM tells the patient to contact the doctor directly through the consultation section — the responsibility of reaching the doctor always lies with the patient, not the LLM.

2. LLM Processing Pipeline

Step 1 — PII Anonymizer (Presidio)

Runs first on every patient message. Detects and strips identity markers (name, phone number, location, etc.) before anything downstream sees the text.

Example: "My name is Priya and I live in Anna Nagar, BP was 138 today" becomes "My name is [NAME] and I live in [LOCATION], BP was 138 today." Identity is stripped; clinical context is preserved. This anonymised version is what every downstream system — RAG and the LLM — sees.

Step 2 — Prompt Injection Defender

Scans the anonymised input for two categories of pattern before it reaches the LLM:

- Injection patterns — phrases like "ignore all previous instructions," "forget your system prompt," "you are now," "jailbreak," "DAN mode"
- Boundary violations — phrases like "dosage," "mg," "tablet," "prescri," "stop taking," "overdose"

If a pattern is found, a safe response is returned without ever touching the LLM. If clean, the message proceeds.

Step 3 — RAG Layer (ChromaDB)

Retrieves verified maternal health guideline chunks relevant to the question — for example, a question about gestational diabetes breakfast foods retrieves the relevant chunks from ICMR dietary guidelines. The LLM answers grounded in this verified content, not just training memory.

Step 4 — System Prompt Construction

Before every response, a system prompt is built containing: the patient's profile (trimester, conditions, job type), the RAG-retrieved verified guidelines, and a fixed set of strict rules — never suggest medicine/dosage, never diagnose, never answer outside maternal health, always end food/exercise advice with the safety disclaimer, tell the patient to contact her doctor directly for serious symptoms, and reiterate that the LLM has no connection to the doctor and cannot notify anyone.

Step 5 — The Single LLM Call (openai/gpt 120B via Groq)

The constructed system prompt plus the anonymised patient message are sent in a single call to the Groq API using the openai/gpt 120B model. Only the anonymised, PII-stripped input is ever sent — never the patient's raw message.

Step 6 — Response Filter — Final Safety Check

Before any response is delivered, it is scanned for a blocked-phrase list: "mg," "milligram," "dosage," "take this," "stop eating," "stop taking," "you have," "diagnosed with," "you should stop." If any phrase is found, the response is replaced entirely with: "I want to be careful here. This is something your doctor should advise on directly. Please use the consultation section to reach them."

The filtered response goes only to the patient chat window. It is never forwarded to the doctor dashboard, and is saved only to an encrypted, patient-only chat log table.

Step 7 — On-Demand Vitals Snapshot — The Blind Vitals Rule

By default the LLM is completely blind to the patient's vitals database — it has no idea what her blood pressure or blood sugar averages are. If the patient clicks "Attach My Current Vitals," the backend queries MySQL for her latest reading (HR, SpO₂, BP, glucose, timestamp), formats it as a plain-text snapshot, and passes it to the LLM for that single message only. After the LLM responds, the snapshot is wiped from context. The doctor dashboard is never informed of this interaction.

3. What the LLM Handles — The 12 Aspects

#	Aspect	What the LLM Does
1	Diet and nutrition	Recommends safe foods based on trimester and health conditions
2	Exercise and physical activity	Suggests safe exercises based on trimester, job type, and conditions
3	Emotional and mental wellbeing	Listens, acknowledges, and provides calming and coping guidance
4	Sleep and rest guidance	Advises on sleep positions, routines, and rest based on trimester
5	Hydration and water intake	Recommends daily fluid targets and practical hydration habits
6	Trimester-specific lifestyle tips	Proactively shares general lifestyle guidance for each trimester

#	Aspect	What the LLM Does
7	Baby kick tracking reminders	Reminds patient to count kicks and guides action based on count
8	General pregnancy questions	Answers questions using a consistent decision framework
9	Report explanation	Translates medical terms — never advises treatment
10	Ask the Doctor question generator	Helps patient form the right questions to ask her doctor
11	On-demand vitals analysis	Analyses vitals — never gives autonomous fix instructions
12	Companion conversations	Warm friendly chat for emotional support and lighthearted moments

Aspect 1 — Diet and Nutrition

Before recommending, the LLM collects trimester, existing conditions (gestational diabetes, high BP, anaemia, thyroid), allergies/dislikes, and vegetarian/non-vegetarian preference. Safe baseline foods include leafy greens, lentils/dal, whole grains, fresh fruits, curd and pasteurised milk, nuts in moderation, cooked eggs, and fresh vegetables — this is a baseline, not a hard limit; the LLM can recommend other pregnancy-safe foods based on context, provided it never guesses on unfamiliar or controversial foods.

Foods it never recommends under any circumstance: raw/unripe papaya, raw pineapple, raw eggs or undercooked meat, unpasteurised milk/cheese, high-mercury fish, excess liver products, unconfirmed herbal teas, alcohol in any form, and excess caffeine (more than one cup per day).

Every food recommendation ends with: "Every pregnancy is different. Always confirm dietary changes with your doctor, especially if you have any existing conditions." When uncertain about any food, the LLM says it is not certain and directs the patient to her doctor rather than guessing. It never suggests supplement dosages or comments on prescriptions, including common items like iron tablets.

Aspect 2 — Exercise and Physical Activity

Before suggesting, the LLM collects trimester, high-risk conditions (placenta previa, pre-eclampsia, preterm labour risk, multiple pregnancy), prior activity level, job type, current pain, and any doctor-imposed restriction.

Trimester	Safe Exercises
1st (weeks 1–12)	Walking 20–30 min daily, gentle prenatal yoga, light stretching, deep breathing
2nd (weeks 13–26)	Brisk walking 30 min, prenatal yoga, stationary cycling, pelvic floor exercises, light stretching
3rd (weeks 27–40)	Slow gentle walking only, pelvic floor exercises, breathing exercises, seated gentle stretching

Never recommended: high-intensity cardio/HIIT, heavy weightlifting, contact sports, hot yoga/sauna, lying flat on back from 2nd trimester onward, jumping/bouncing, or any activity with fall risk. The LLM personalises by job type (e.g., a desk-job patient gets movement breaks every 45 minutes; a physical-job patient is advised to focus on rest and stretching instead of added exercise) — these are illustrative scenarios, not an exhaustive list.

The LLM tells the patient to stop exercise immediately and contact her doctor if she reports vaginal bleeding, chest pain, palpitations, sudden shortness of breath, dizziness, painful contractions, or reduced baby movement during/after activity. For any patient with a high-risk condition, it does not suggest exercise at all and instead asks her to get specific clearance from her doctor.

Aspect 3 — Emotional and Mental Wellbeing

The LLM is a warm, always-available listening presence — not a therapist. It applies an acknowledge-first, listen-second, respond-third approach to every emotional message, and opens each chat session with a mood check-in ("How are you feeling today, physically and emotionally?"). For anxiety or stress it guides a simple breathing exercise (in for 4, hold for 4, out for 6).

Trimester	Common Emotional Issue	LLM Focus
1st	Anxiety and nausea-related frustration	Reassurance and normalising symptoms
2nd	Body image and relationship changes	Positive affirmations and gentle encouragement
3rd	Delivery fear and sleep anxiety	Breathing techniques and birth preparation tips

The LLM never diagnoses depression or anxiety disorders, never provides therapy, and never handles self-harm or suicidal ideation clinically. If a patient expresses self-harm, suicidal thoughts, or severe distress, it responds: "I am very concerned about what you have shared. Please speak to your doctor immediately or contact a mental health professional. Please use the consultation section to reach your doctor right now," and does not go further into the topic.

Aspect 4 — Sleep and Rest Guidance

Collects trimester, back/pelvic pain, night-shift work, and breathing difficulties/snoring before advising. Key guidance: 1st trimester allows daytime napping and no position restriction with 8–10 hours recommended; 2nd trimester introduces left-side sleeping and avoids lying flat on the back; 3rd trimester makes left-side sleeping mandatory with pillow support and back-lying fully avoided. General hygiene tips: fixed sleep/wake time, no screens 30 minutes before bed, light dinner, cool dark room, and a short 20-minute afternoon nap if needed.

The LLM directs the patient to her doctor — without diagnosing — for heavy snoring, multi-day inability to sleep, or severe restless legs.

Aspect 5 — Hydration and Water Intake

Recommends 2.5–3 litres of total fluid per day (including coconut water, buttermilk, dal, water-rich fruits), with plain water preferred for diabetic patients. It gives a practical routine (2 glasses on waking, 1 before each meal, sipping between meals, 1 before bed) rather than asking the patient to count litres. It advises avoiding caffeinated drinks beyond one cup, carbonated drinks, packaged juices, and very cold water. It directs the patient to her doctor for dark yellow urine, persistent headache, dizziness, no urination for 6+ hours, or intermittent contractions.

Aspect 6 — Trimester-Specific Lifestyle Tips

Delivered proactively as soon as the trimester is known, without the patient needing to ask, 2–3 tips per conversation rather than dumped all at once. Examples span folic acid reminders (without specifying dosage), avoiding alcohol/smoking/raw foods, left-side sleep habit-building, kick tracking, dental checkups, footwear, hospital-bag packing by week 36, and knowing the route to the hospital in advance. These are a strong baseline, not an exhaustive ceiling — the LLM shares additional relevant tips contextually.

Aspect 7 — Baby Kick Tracking Reminders

The clinical minimum is 10 kicks per day (not per 2 hours), best counted after a meal while lying on the left side. The LLM sends a daily reminder and responds based on the count reported:

Count Reported	LLM Response
10+ kicks in the day	Reassures — "Baby is active and responding well today."

Count Reported	LLM Response
Fewer than 10 so far	Suggests a cold drink/snack, lying on left side, and counting again later
Still fewer than 10 by end of day	"Please contact your doctor immediately. Do not wait."
No movement at all today	Tells patient to contact doctor immediately, without delay

Every count is logged with patient ID, date, kicks counted, time taken, and flag status — visible to the doctor on the patient dashboard independently, but never pushed to the doctor by the LLM itself. The LLM never contacts the doctor on the patient's behalf; the patient must act.

Aspect 8 — General Pregnancy Questions (Decision Framework)

Rather than pre-programmed FAQ answers, every unpredictable question runs through a consistent 4-step framework:

- Step 1 — Does answering require her specific medical history? If yes, answer generally and add a doctor-confirmation note.
- Step 2 — Is the answer medically safe and evidence-based? If not sure, redirect to doctor.
- Step 3 — Does the question involve a symptom she is experiencing? If yes, take it seriously, ask a follow-up, never dismiss.
- Step 4 — Does it involve medicine, diagnosis, or a clinical decision? If yes, redirect to doctor immediately, no exceptions.

Three golden rules apply to every question: never dismiss a question as silly, when in doubt redirect to the doctor rather than risk a wrong reassurance, and never make the patient feel stupid regardless of how basic the question seems.

Aspect 9 — Report Explanation

Acts strictly as a medical translator, converting lab/scan terminology into plain language — never advising treatment. It is allowed to translate terms, explain what a value means generally, and state whether a value is above/below/within a commonly referenced range. It is strictly blocked from saying what the patient should do about a result, suggesting food/supplement/medicine in response, diagnosing, or comparing reports to draw clinical conclusions.

NOTE: Hard guardrail: every report explanation must end with the exact line — "For what this means for your health and any steps to take, please consult your doctor directly through the consultation section." This is non-negotiable.

Aspect 10 — "Ask the Doctor" Question Generator

When the patient attaches vitals or uploads a report and asks what to ask her doctor, the LLM reads the data, identifies out-of-range readings, and generates a 3–5 bullet question script for her to bring to the doctor.

NOTE: Hard guardrail: the LLM generates questions for the doctor — it never answers those questions itself. It says "ask your doctor about your BP of 135," never "your BP of 135 means you should do X."

Aspect 11 — On-Demand Vitals Analysis

When vitals are attached, the LLM may describe what it sees (above/below/within range, different from recent trend) but is strictly blocked from saying what to eat/drink/do to fix a reading, suggesting rest or medication, reassuring that an elevated reading is fine, or making any clinical severity judgement.

NOTE: Hard guardrail: every vitals analysis involving an out-of-range reading must end with — "Please open the Consultation Section on this platform to reach your doctor directly." The LLM intercepts any attempt to get a fix or action plan and redirects. No exceptions.

Aspect 12 — Companion Conversations

When the patient is in a casual, lighthearted mood rather than seeking medical information, the LLM drops its clinical tone completely and becomes a warm, friendly companion — for example playing simple word games (like a baby-name alphabet game), discussing nursery themes or baby names, or just chatting about books, movies, or cravings at 2 AM when the patient is bored or can't sleep.

Even in this mode, the LLM never gives medical advice, never discusses medicines/dosages/symptoms casually, and never pretends to be a doctor or makes clinical statements. It is handled by the same single LLM pipeline as every other conversation — the tone shift is driven by the system prompt and conversational context, not a separate model.

4. What the LLM Never Responds To

For all eight items below, the LLM responds with: "Please contact your doctor directly through the consultation section," and does not attempt to answer, guess, or partially help.

#	Aspect	Why
1	Prescription queries	Prescriptions are a clinical decision made by the doctor
2	Medicine names or dosages	Wrong dosage information during pregnancy can be life-threatening
3	Changing or stopping medication	Only the prescribing doctor can modify or discontinue a medicine
4	Diagnosing symptoms	Diagnosis requires examination and clinical judgement, beyond LLM scope
5	Lab report/scan treatment advice	The LLM translates terms but never advises next steps
6	Emergency medical situations	The LLM tells the patient to go to hospital immediately; it does not manage emergencies
7	Surgical or clinical decisions	Entirely outside the LLM's role, always a doctor decision
8	Autonomous fix instructions for vitals	The LLM analyses and redirects to consultation, never tells the patient how to fix a reading

5. General User Behaviour

A general (non-maternal) user chats about general topics — stocks, sales, personal interests, everyday questions — and the LLM responds freely to non-health topics. For any health-related question, it shows: "This assistant is designed for maternal health guidance only." General users cannot bypass maternal health restrictions regardless of how the question is framed.

6. Notifications System

Patient-Side Notifications

#	Notification Type	What Triggers It	Delivery Method
1	Critical vitals alert	Her own vitals cross a danger threshold — BP above 140, SpO ₂ below 94, HR above 120, glucose below 70 or above 140	Push notification + SMS, both sent simultaneously
2	Doctor message	Doctor sends a message in the tele-consultation chat	Push notification
3	Appointment reminder	An appointment is approaching	Push notification 1 hour before — SMS also sent
4	Daily kick count reminder	She has not logged her kick count for the day	Push notification once daily in the afternoon
5	Medicine order status	Order placed, dispatched, or delivered	Push notification at each status change
6	New report uploaded	Hospital uploads a blood test, scan, or other report	Push notification with report type and date

How Push Notifications Work

Browser push notifications via the Web Push API work even when the browser is closed or minimised. On first login the browser asks: "Allow notifications from this app?" — the patient clicks Allow. After that, notifications arrive automatically without the app being open, and clicking any notification opens the app directly to the relevant section (e.g., a new-report notification opens directly to the My Reports section under that hospital).

Critical Vitals Alert — Special Handling

This is the most important notification type, delivered as both push notification and SMS simultaneously so the patient receives it even if her phone is on silent or the app is not installed on that device. Flow: BP reads 155/98 → backend detects the threshold breach (systolic > 140) → push notification sent ("Your blood pressure reading is elevated at 155/98 mmHg. This is above the safe threshold of 140/90. Please contact your doctor immediately.") → SMS sent simultaneously → if she has a connected doctor with toggle ON, he receives the alert on his dashboard at the same moment.

NOTE: The LLM does NOT trigger or manage these notifications. Vitals alerts are handled entirely by the backend threshold detection system (built in Phase 3). The LLM has no connection to the notification pipeline.

Doctor-Side Notifications

Notification Type	What Triggers It	Delivery Method
Critical vitals alert	A patient with toggle ON crosses a danger threshold	Push notification + red banner, simultaneous with the patient's own alert
New patient assigned	A hospital maps a new patient to him at an already-unlocked hospital	Push notification
Access deactivated	A hospital deactivates his profile there	Push notification and banner inside that hospital's section
Tele-consultation message	A patient sends him a message in the shared chat	Push notification

Hospital-Side Notifications

Notification Type	What Triggers It	Delivery Method
New patient connected	A patient successfully enters her access code	Push notification on the hospital dashboard

Notification Type	What Triggers It	Delivery Method
Appointment booked	A patient books an available slot with one of the hospital's doctors	Push notification
Critical vitals alert (patient side)	A connected patient's vitals cross a danger threshold with sharing toggle ON for that doctor	Delivered to the doctor's own dashboard — the hospital account itself is not separately alerted, consistent with per-doctor scoping
Access code expiring soon	3 days remain before a generated code's 30-day validity ends, still unused	Push notification reminding staff to hand it over or regenerate

Phase 4 Summary Table

Component	Key Rule Established
LLM model	Single model — openai/gpt 120B via Groq API. No routing agent, no second local model.
Security pipeline	PII Anonymizer → Prompt Injection Defender → RAG (ChromaDB) → LLM call → Response Filter.
Doctor isolation	LLM output never reaches the doctor dashboard; stored only in an encrypted patient-only chat log.
Blind vitals rule	LLM has no standing database access; snapshot is attached on demand and wiped after use.
12 supported aspects	Diet, exercise, emotional support, sleep, hydration, tips, kicks, general Q&A, report explanation, ask-doctor generator, vitals analysis, companion chat.
8 hard refusals	Prescriptions, dosages, medication changes, diagnosis, report treatment advice, emergencies, surgical decisions, autonomous vitals fixes.
Notifications	Patient (6 types), Doctor (4 types), Hospital (4 types) — push via Web Push API, SMS for critical events.